

120 Days of Quantitative Finance

Month 4: Stochastic Calculus & Time Series Modeling

Week 14: SDEs and Black-Scholes

Day 95: Derivation of the Black-Scholes Equation

1 Introduction

The Black-Scholes equation is a fundamental result in financial mathematics used to price derivative securities.

It is derived using stochastic calculus, Ito's Lemma, and the concept of risk-neutral replication.

2 Stock Price Dynamics

Assume the stock price follows Geometric Brownian Motion:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

3 Option Price Function

Let the option price be:

$$V = V(S, t)$$

4 Applying Ito's Lemma

$$dV = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} (dS)^2$$

Substituting:

$$dS = \mu S dt + \sigma S dW_t$$

and using:

$$(dW_t)^2 = dt$$

we get:

$$dV = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S} dW_t$$

5 Constructing a Risk-Free Portfolio

Define portfolio:

$$\Pi = V - \Delta S$$

Choose:

$$\Delta = \frac{\partial V}{\partial S}$$

Then stochastic term cancels:

$$d\Pi = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt$$

6 No-Arbitrage Argument

Since the portfolio is risk-free:

$$d\Pi = r\Pi dt$$

Substituting:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

7 Black-Scholes PDE

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

8 Interpretation

- No dependence on μ (real-world drift)
- Pricing depends only on risk-free rate r
- Eliminates risk via hedging

9 Conclusion

The Black-Scholes equation is derived using:

- Ito's Lemma
- dynamic hedging
- no-arbitrage principle

It forms the foundation of modern option pricing theory.